

Exotics

(2. Properties of American Option)

American Option (Recap)

Consider a binomial tree with n periods. Define the payoff of the security as G_k which is what the holder will get if he exercises it at time k

Because the holder of an American option has the right to exercise the option at any time, we defined the binomial pricing function as

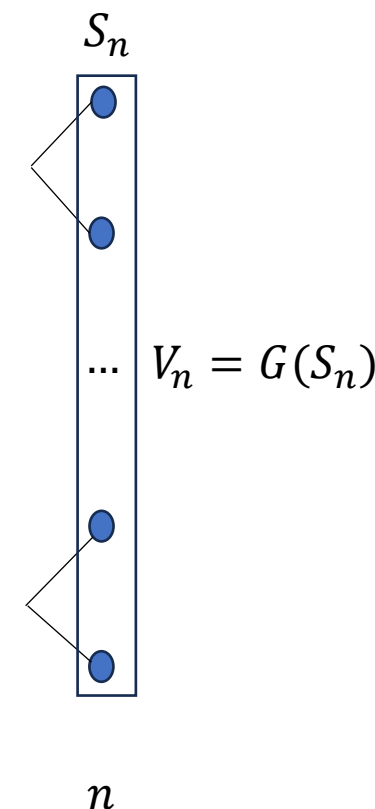
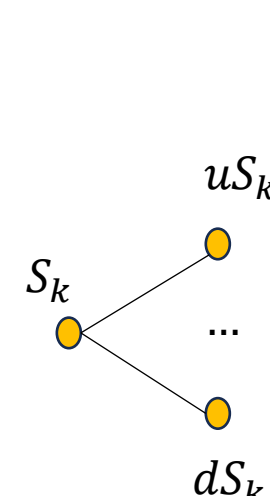
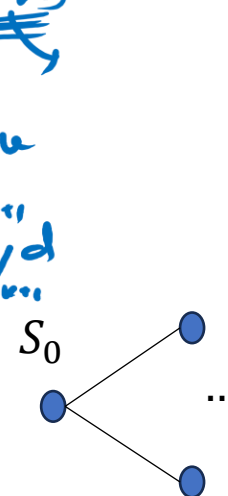
$$\begin{cases} V_k = G_n, \\ V_k = \max \left(\underbrace{G_k}_{\text{(Exercise Now)}}, \underbrace{\mathbb{E}^{\mathbb{Q}} \left(B_k \frac{V_{k+1}}{B_{k+1}} \middle| \mathcal{F}_k \right)}_{\text{(Continue)}} \right), \end{cases}$$

$$G_k = (S_k - K)^+$$

$$\begin{aligned} k &= n \\ k &< n \end{aligned}$$

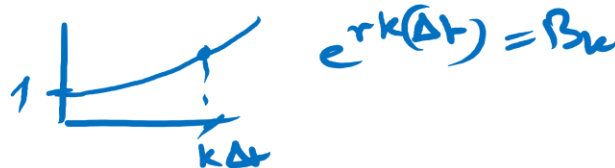
$$(S - K)^+$$

(Binomial Tree for Stock)



Discounted Quantities

Define the discounted quantities



$$\tilde{V}_k = \frac{V_k}{B_k} \quad (\text{discounted value process})$$

$$\tilde{G}_k = \frac{G_k}{B_k} \quad (\text{discounted payoff process})$$

The we can redefine the recursive formula as

$$\tilde{V}_k = \begin{cases} \tilde{V}_k = \tilde{G}_n, & k = n \\ \tilde{V}_k = \max(\tilde{G}_k, \mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1} | \mathcal{F}_k)), & k < n \end{cases}$$

$$V_k \geq G_k$$

The above process $\tilde{V}_k$ is called the **Snell Envelope** of the process $\tilde{G}_k$.

Theorem (Snell Envelope)

Theorem 1 : The Snell Envelope $\tilde{V}_k$ is the smallest super-martingale that dominates the process $\tilde{G}_k$

Clearly, $\tilde{V}_k = \max(\tilde{G}_k, \mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1} | \mathcal{F}_k))$ suggests that $\tilde{V}_k \geq \tilde{G}_k$ (dominant)

Also, the above suggests $\tilde{V}_k \geq \mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1} | \mathcal{F}_k)$ which means it's a super-martingale (trending down)

The fact that it is smallest can be proved using induction.

$$\tilde{V}_k \geq \mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1} | \mathcal{F}_k)$$

Stopping Time (Recap)

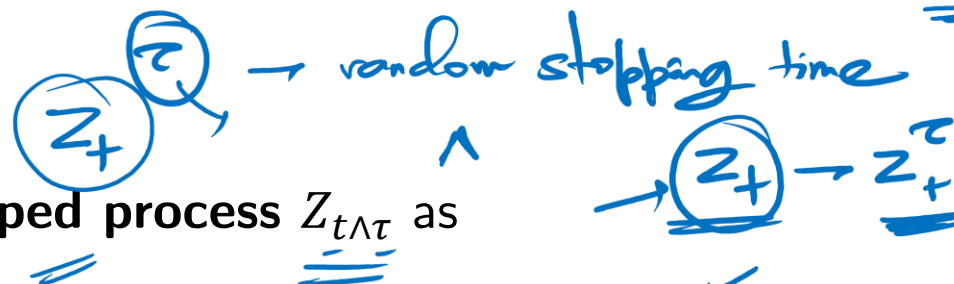
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the probability space with filtration $\{F_k\}_{k=0}^n$. A **stopping time** is a random variable which is defined as $\tau: \Omega \rightarrow \{0, 1, 2, \dots, n\} \cup \infty$ with the property that

$$\{\omega \in \Omega; \tau(\omega) \leq k\} \in \mathcal{F}_k, \forall k = 0, 1, 2, \dots, n, \infty$$

Which means that one only needs observations till k , to confirm if the stopping time $= k$ and need not look into the future. In American pricing context, the early exercise strategy is **admissible** if it's a stopping time.

We also defined a **stopped process** $Z_{t \wedge \tau}$ as

$$Z_t^\tau = Z_{t \wedge \tau} = \begin{cases} Z_t, & t \leq \tau \\ Z_\tau, & t > \tau \end{cases}$$



Stopped process and martingale property

✓ **Theorem 2** : If the original process Z_t is a (super/sub) martingale then the stopped process Z_t^τ is also a (super/sub) martingale

Using the above theorem, we can conclude immediately that $\tilde{V}_k^\tau$ is a super-martingale

$\tilde{V}_k \rightarrow \text{super-martingale}$

Theorem 3 : Let's define our stopping time τ (buyer's exercise policy) as

$$\tau = \min_k \{V_k = G_k\}$$

$\frac{V}{B} = \frac{G}{B}$

Then the $\tilde{V}_k^\tau$ is a $\mathbb{Q}$ -Martingale (**stronger condition**)

→ (Note that this stopping time is when the Snell envelope first time meets the payoff)

Proof

Remember $\tilde{V}_k = \max(\tilde{G}_k, \mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1} | \mathcal{F}_k))$. Under the defined stopping time, if $\tilde{V}_k > \tilde{G}_k$ (no exercise) then

$$\underbrace{k} \quad \underbrace{k+1} \quad \underbrace{\tilde{V}_k = \mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1} | \mathcal{F}_k)}$$

Now let's focus on the increments of the stopped process $\tilde{V}_k^\tau$. The change in this process is given as

$$\tilde{V}_{k+1}^\tau - \tilde{V}_k^\tau = \begin{cases} 0, & k+1 > \tau \\ \tilde{V}_{k+1} - \tilde{V}_k, & k+1 \leq \tau \end{cases} = \begin{cases} 0, & k+1 > \tau \\ \tilde{V}_{k+1} - \mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1} | \mathcal{F}_k), & k+1 \leq \tau \end{cases}$$

Taking conditional expectation $\mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1}^\tau - \tilde{V}_k^\tau | \mathcal{F}_k) = 0$ (no drift)

Hence, $\tilde{V}_k^\tau$ is a $\mathbb{Q}$ -Martingale

$$\tau \rightarrow$$

$$\mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1} | \mathcal{F}_k) - \mathbb{E}^{\mathbb{Q}}(\tilde{V}_{k+1} | \mathcal{F}_k)$$

More Theorems (I)

Theorem 4 : The value of American pricing formula is given as

$$V_0 = \max_{\tau} \left\{ \mathbb{E}^{\mathbb{Q}} \left(\frac{G_{\tau}}{B_{\tau}} \middle| \mathcal{F}_0 \right) \right\}$$

Handwritten annotations: A circled τ with an arrow pointing to the max operator. A circled V_0 with an arrow pointing to the max operator. A circled $\frac{G_{\tau}}{B_{\tau}}$ with an arrow pointing to the expectation operator. A circled τ with an arrow pointing to the max operator.

Where the maximum is taken over all stopping times τ (permissible strategies). It intuitively, makes sense as the holder would want to maximize expected wealth by choosing the optimal policy.

Theorem 5 : Any stopping time τ which satisfies $V_0 = \mathbb{E}^{\mathbb{Q}} \left(\frac{G_{\tau}}{B_{\tau}} \middle| \mathcal{F}_0 \right)$ is optimal.

For example, $\tau = \min_k \{V_k = G_k\}$ is optimal.

Handwritten annotations: A circled V_0 with an arrow pointing to the expectation operator. A circled $\frac{G_{\tau}}{B_{\tau}}$ with an arrow pointing to the expectation operator. A circled $\tau = k$. A circled $\tau(\omega) = k$. A circled τ with an arrow pointing to the min operator. A circled τ with an arrow pointing to the min operator.

More Theorems (II)

Theorem 5 : Doob Decomposition of Super Martingales

Let U_n be the Snell envelope (super-martingale) of X_n .
There exists a unique decomposition such that

- 1) M_n is a martingale
- 2) A_n is a decreasing predictable process $\rightarrow$ previsible

Proof is straightforward,

$$U_{n+1} - U_n = (M_{n+1} - M_n) + (A_{n+1} - A_n)$$

$$\mathbb{E}(U_{n+1} - U_n | \mathcal{F}_n) = (A_{n+1} - A_n)$$

$$A_{n+1} \leq A_n \rightarrow \text{decreasing}$$

$$U_n = M_n + A_n$$

$$X_0 + \sum_{k=1}^n (X_k - \mathbb{E}[X_k | \mathcal{F}_{k-1}])$$

(adds up the surprises)

$$\sum_{k=1}^n (\mathbb{E}[X_k | \mathcal{F}_{k-1}] - X_{k-1})$$

(adds up the expected increments)

Then the largest optimal stopping time. It gives the last moment before the discounted value of the American option will drop in expectation; up to time $\tau_{\max}$ the discounted value process U is a martingale with respect to Q

$$\tau_{\max} := \begin{cases} N & \text{if } A_N = 0, \\ \min\{n \in \{0, \dots, N-1\} \mid A_{n+1} < 0\} & \text{if } A_N < 0. \end{cases}$$

American Call

1) Show that American call option should never be exercised prior to maturity.

Let the discounted price of American and European call option be C and c respectively.

At expiry, we have

$$\underline{C_T = c_T = G_T}$$

Since C is a super-martingale and c is a martingale,

$$C_t \geq \mathbb{E}^{\mathbb{Q}}(C_T | \mathcal{F}_t) = \mathbb{E}^{\mathbb{Q}}(c_T | \mathcal{F}_t) = c_t$$

Since $c_t \geq S_t - Ke^{-r(T-t)}$, it also satisfies $c_t \geq S_t - K$, which means the European call option price is always larger than the intrinsic value (time value > 0). Therefore,

$$C_t \geq S_t - K$$

Hence, the American call should be continued and never exercised early.

American Put



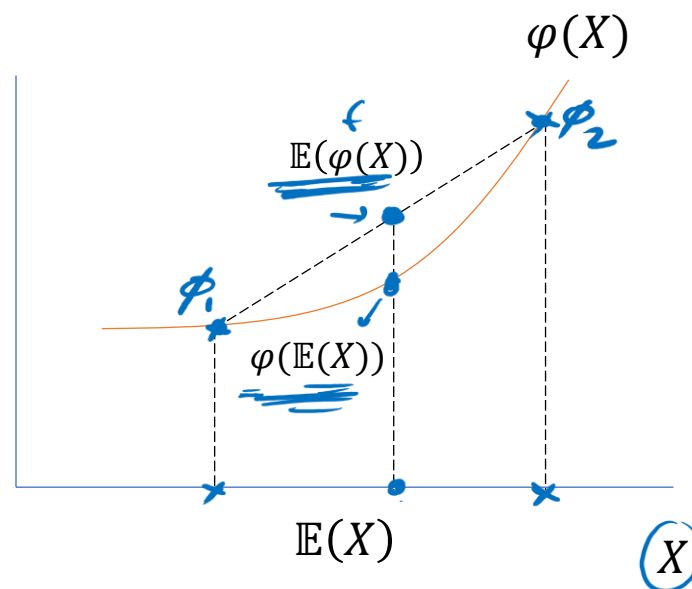
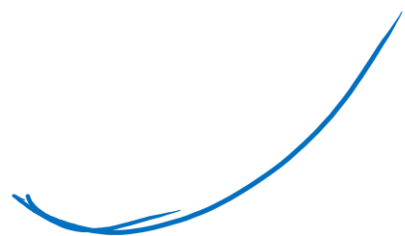
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(Think!)

Jensen's inequality

If $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is convex, and $\mathbb{E}|\varphi(X)| < \infty$ the

$$\mathbb{E}(\varphi(X)|\mathcal{F}) \geq \varphi(\mathbb{E}(X|\mathcal{F}))$$



Jensen's inequality (I)

Theorem 6 : If Y_k is a martingale and φ is convex, then $\varphi(Y_k)$ is a sub-martingale.

Proof :

$$\begin{aligned} \rightarrow \mathbb{E}(\varphi(Y_{k+1})|\mathcal{F}_k) &\geq \varphi(\mathbb{E}(Y_{k+1}|\mathcal{F}_k)) \\ &= \varphi(Y_k) \end{aligned}$$

Jensen's inequality (II)

The payoff function $g(x) = (x - K)^+$ for an American call is a Convex function, so we have $g(\lambda x) \leq \lambda g(x)$ ✓

Hence

$$g\left(B_k \frac{S_{k+1}}{B_{k+1}}\right) \leq \frac{B_k}{B_{k+1}} g(S_{k+1})$$

$\frac{g(S)}{B}$

Also

$$\begin{aligned} \mathbb{E}\left(\frac{g(S_{k+1})}{B_{k+1}} \middle| \mathcal{F}_k\right) &= \frac{1}{B_k} \mathbb{E}\left(B_k \frac{g(S_{k+1})}{B_{k+1}} \middle| \mathcal{F}_k\right) \\ &\geq \frac{1}{B_k} \mathbb{E}\left(g\left(B_k \frac{S_{k+1}}{B_{k+1}}\right) \middle| \mathcal{F}_k\right) \geq \frac{1}{B_k} g\left(\mathbb{E}\left(B_k \frac{S_{k+1}}{B_{k+1}} \middle| \mathcal{F}_k\right)\right) = \frac{g(S_k)}{B_k} \end{aligned}$$

So, the expected discounted payoff of an American Call is a submartingale, meaning the waiting value is higher than the exercise value. Therefore, American call should never be exercised early.

